



Operational Best Practices

ACE Team

CoPilot Remote Syslog

- The Syslog server is by default the CoPilot.
 - ❑ When you deploy the CoPilot, automatically the Controller will open the port UDP **5000** (Syslog) on the Security Group attached to the CoPilot instance.
 - ❑ Aviatrix CoPilot Remote Syslog Profile is set to use the Profile **9**.

Remote Syslog

✔ Enabled

Aviatrix CoPilot Remote Syslog Profile

Server	[REDACTED]
Protocol	UDP
Port	5000

Edit Profile

- In the **CoPilot > Settings > Configuration > Logging Services** page, you can configure the forwarding of logs from the Aviatrix platform to the log servers of well-known log management systems.

Manage Remote Syslog

Profile

Profile 2 × ▾

Profile Name

SolarWinds' Kiwi Syslog

Server

[REDACTED]

Protocol

UDP ▾

Port

5000

▼ Certificate

▼ Custom Template

▼ Advanced Settings

Cancel

Save

CoPilot NetFlow



- The NetFlow collector is by default the CoPilot.
 - ❑ When you deploy the CoPilot, automatically the Controller will open the port UDP **31283** (NetFlow) on the Security Group attached to the CoPilot instance.
 - ❑ All the Aviatrix Gateways will send NetFlow data to the CoPilot.
 - ❑ As a consequence, FlowIQ feature in Aviatrix CoPilot will start to process the NetFlow information received by the Gateways.

Netflow Agent ✔ Enabled

Server	[REDACTED]
Port	31283
Mode	IPT Mode

[Edit Configuration](#)

- In the **CoPilot > Settings > Configuration > Logging Services** page, you can configure the forwarding of the NetFlow Data from the Aviatrix platform to your designated service point.

Manage Netflow Agent

Repository

Server: [REDACTED]

Port: 31283 Version: 9

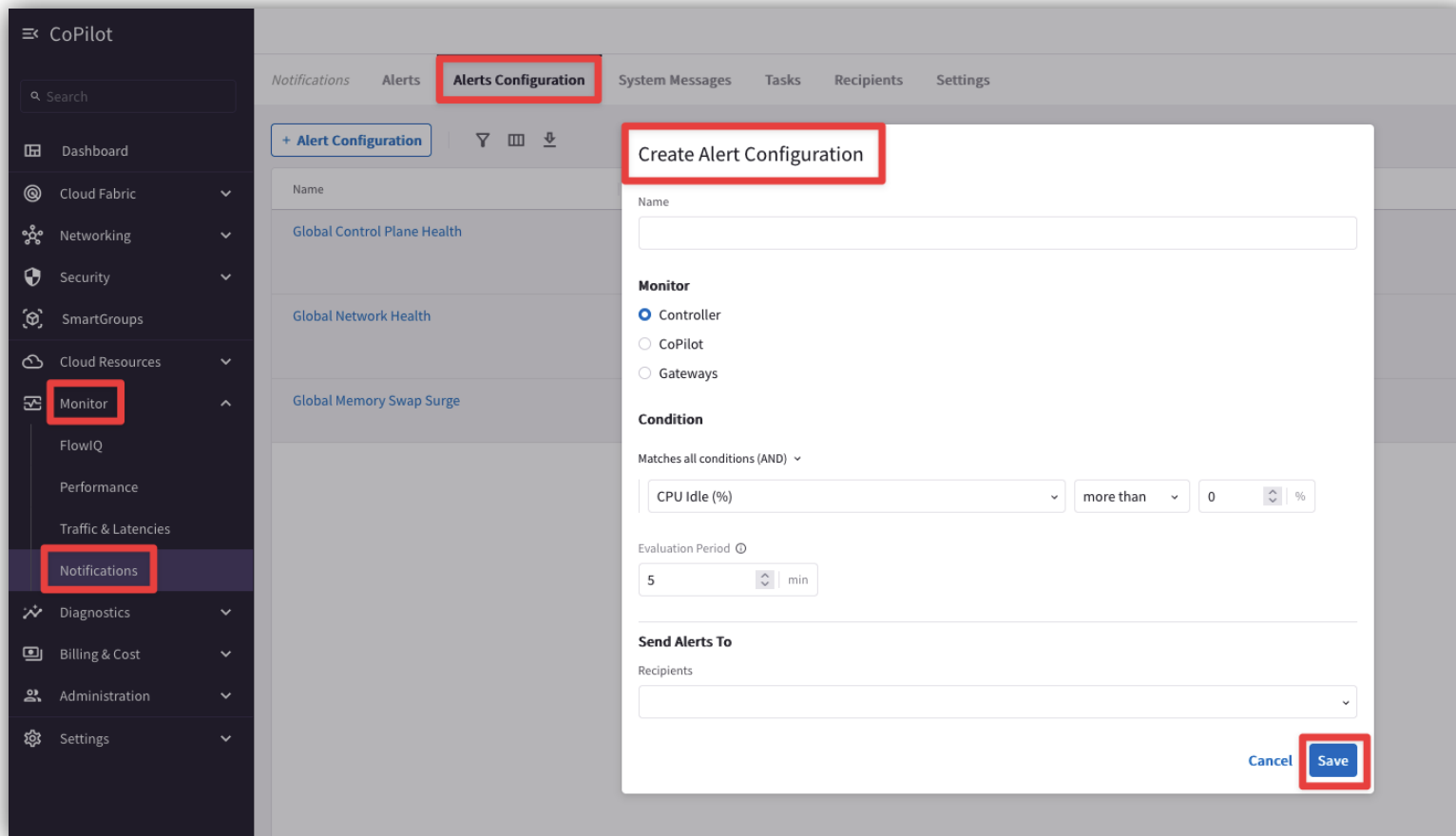
Mode:
☒ IPT
☐ L7 Preview Feature

▼ Advanced Settings

[Disable](#) [Cancel](#) [Save](#)

CoPilot Alerts Configuration

1. Webhooks Integrations work with any 3rd party integration (Slack, PagerDuty, ServiceNow, etc.)
2. Add webhook endpoints (can send payload as JSON or text)
3. Provide custom tags in the payload to classify triggered events and further integrate into your systems
4. Get alerted via webhook and email for the same alert



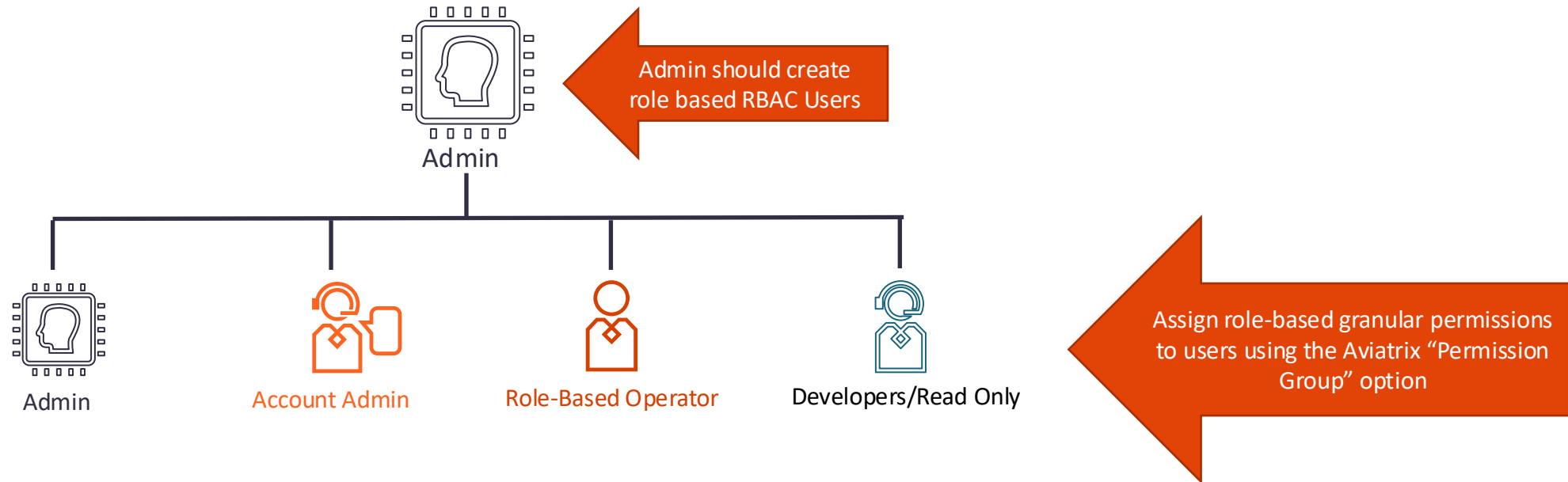
The screenshot displays the CoPilot Alerts Configuration interface. The left sidebar contains a navigation menu with the following items: Dashboard, Cloud Fabric, Networking, Security, SmartGroups, Cloud Resources, Monitor (highlighted with a red box), FlowIQ, Performance, Traffic & Latencies, Notifications (highlighted with a red box), Diagnostics, Billing & Cost, Administration, and Settings. The main content area has tabs for Notifications, Alerts, Alerts Configuration (highlighted with a red box), System Messages, Tasks, Recipients, and Settings. Below the Alerts Configuration tab is a '+ Alert Configuration' button and a list of existing alerts: Global Control Plane Health, Global Network Health, and Global Memory Swap Surge. A 'Create Alert Configuration' modal is open, featuring a 'Name' field, a 'Monitor' section with radio buttons for Controller (selected), CoPilot, and Gateways, a 'Condition' section with a dropdown set to 'Matches all conditions (AND)' and a rule for 'CPU Idle (%)' set to 'more than' 0%, an 'Evaluation Period' of 5 minutes, and a 'Send Alerts To' section with a 'Recipients' dropdown. The modal includes 'Cancel' and 'Save' (highlighted with a red box) buttons at the bottom right.



Role-Based Access Control (RBAC)

RBAC: Role-Based Access Control

Aviatrix RBAC is based on the “Users” and “Permission Group” concept



RBAC with SAML/OKTA

<https://www.youtube.com/watch?v=l9zmZTaVUo8>

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User Access- CoPilot



☰ CoPilot

Dashboard

Cloud Fabric

Networking

Security

Groups

Cloud Resources

Monitor

Diagnostics

Administration

User Access

Reports

User AccessUsersPermission GroupAccess Management

+ User

Name	Email
admin	ace.lab@aviatrix.com
copilot_service_account	ace.lab@aviatrix.com
johndoe	johndoe@aviatrix.com
student	ace.lab@aviatrix.com

Add User

Username

cloudE-engineer

Email

engineer@aviatrix.com

Password

.....

👁

Confirm Password

.....

👁

Permission Groups

security ×

× ▾

Cancel

Save

Permission Group (part.1)



☰ CoPilot

🔍 Search

Dashboard

Cloud Fabric

Networking

Security

Groups

Cloud Resources

Monitor

Diagnostics

Administration

User Access

Reports

User AccessUsersPermission GroupAccess Management

+ Permission Group

🔍📄📥

Name	Copilot Visibility	Controller Permissions
admin	All	Write
read_only	All	Read
copilot_permission	All	Write
security-team	Security Groups	

Permission Group (part.2)



Create Permission Group ?

Name

Users

johndoe ×

Access Accounts

aws-account ×

CoPilot Visibility

Controller Permissions

⚠ CoPilot Visibility is in Preview. Preview features are not safe for deployment in production environments. [Learn More](#)

Select All Views

Clear All Views

Search and Select ▼

☑ Cloud Fabric

☑ Topology

5/5 Tabs

All Tabs ×

☑ Gateways

6/6 Tabs

All Tabs ×

☑ Hybrid Cloud

4/4 Tabs

All Tabs ×

☑ Scaling

7/7 Tabs

All Tabs ×

Cancel

Save

☑ Cloud Fabric

☑ Networking

☑ Security

☑ Groups

☑ Cloud Resources

☑ Monitor

☑ Diagnostics

☑ Administration

☑ Settings

Cancel

Save

 aviatrix

9

Authentication Phase



- Users can be authenticated:
 - **Locally** on the Aviatrix Controller
 - Onboard Users (Admin, Operators, Developers, Read-Only)
 - Allowed to reset their password
 - Using **SAML IDP**
 - Onboard Users (Admin, Operators, Developers, Read-Only)
 - Other functionality depends on IDP



AWS SSO



SAML Integration Example – Identity Provider

RBAC User: developer

RBAC User: Admin

RBAC User: Account_A-B

RBAC User: SecOps

- read_only
- Super-Users
- Account-Admin
- Account Admins (A&B)
- Account Admins (C&D)

Security-Users

RBAC-User	Permissions
developer	Read Only
Admin	Super User (Admin)
Account_A-B	CSP Account Admin for Accounts A&B Only
SecOps	Security User



Okta login form with fields for Username, Password, and a Sign In button. It also includes a 'Remember me' checkbox and an Aviaatrix ACE logo in the top right corner.

Aviaatrix login form with fields for Username, Password, and a SIGN IN button. It also includes a 'Forgot password?' link, an OR separator, a dropdown menu with 'aviaatrix_saml_controller' selected, and a highlighted SIGN IN WITH SAML button.

- Admin/Super-Users Admin
- Account Admins Account-A&B
- Security-User SecOps
- Developers/Read Only Developer

Configuring SAML Authentication



User can be authenticated using the local database or using the SAML Integration.

Go to *CoPilot > Administration > User Access > Access Management*

Under Login Authentication, click **+SAML Endpoint**

The screenshot shows the Aviatrix CoPilot interface. On the left, a dark sidebar contains a search bar and a list of navigation items: Dashboard, Cloud Fabric, Networking, Security, Groups, Cloud Resources, Monitor, Diagnostics, Administration, User Access (highlighted with a red box), Reports, Audit, Upgrade, Billing, and Settings. The main content area is titled 'Access Management' and has tabs for 'User Access', 'Users', 'Permission Group', and 'Access Management'. Under 'Access Management', there are two sections: 'Security and Password Settings' and 'Login Authentication'. The 'Login Authentication' section is expanded, showing a 'SAML' subsection with a '+ SAML Endpoint' button highlighted by a red box and a red arrow. Below this is a table with columns: Name, Permission Group, Access Set By, and Test. The table contains one entry: 'aviatrix_saml_sso' with 'profile_attribute' in the 'Access Set By' column and a URL in the 'Test' column.

The 'Create SAML Endpoint' form is shown. It includes the following fields and options:

- Name:** Octa
- SAML Endpoint Configuration:** ☒
- Identity Provider Metadata Type:** ☒ URL ☐ Text
- Identity Provider Metadata URL:** 1Y00006RYxeSAD/with-okta-admin-changes-how-do-i-find-identity-provider-metadata-url?language=en_US
- Entity ID:** ☒ Hostname ☐ Custom
- Access Set By:** ☒ Controller ☐ SAML Identity Provider Attribute
- Permission Group:** (empty dropdown)
- Sign Auth Requests:** ☐ No
- Buttons:** Cancel, Save



Gateway Best Practices

Gateway Deployment – Best Practices

Aviatrix Gateway Optimization & Resiliency

● High Availability (HA):

- **Production:** Deploy multiple gateways across different **Availability Zones (AZs)** for redundancy and automatic failover.
- **Non-Production:** Enable **Single AZ HA** (default in Controller/CoPilot; requires flag in Terraform) to maintain uptime.

● Gateway Sizing & Scaling:

- **Auto Right-Sizing:** Adjust gateway instances based on performance recommendations.
- **Gateway Scaling:** Use for scheduled scaling or adding instances (visible only if configured prior to CoPilot 4.15).
- **Recommendations:** Align instance sizes with your specific bandwidth, VPC, and performance requirements.

● Resiliency & Tuning:

- **Keepalive Timers:** Adjust based on your specific HA recovery objectives.
- **TCP MSS:** Recommended maximums are **1370 bytes** (AWS, Azure, OCI) and **1330 bytes** (GCP).
- **IPsec Anti-Replay:** The default window size of 1 is generally sufficient.



Hub&Spoke or Not? Best Practices

Hub & Spoke Architecture – Best Practices

Multicloud Transit Best Practices

- To ensure a resilient and scalable network architecture, adhere to these core design principles:
- **Centralized Hub-and-Spoke:** Implement a transit network to consolidate connectivity, simplify management, and enforce centralized control across all VPCs and VNets.
- **Regional Redundancy:** Architect transit environments to span at least two cloud regions. Ensure your application stack is configured to maintain operations during a single regional outage.
- **Multicloud Connectivity:** Utilize **MultiCloud Transit Peering** to seamlessly link different cloud providers into a unified transit fabric.
- **Multi-Tier Scalability:** In large-scale deployments, utilize a multi-tier transit architecture to reduce peering complexity and optimize traffic flow.



Aviatrix Controller High Availability (HA)

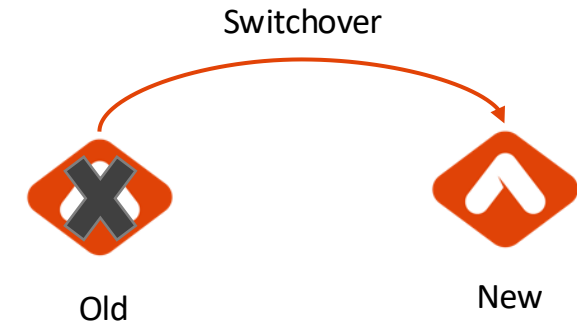
Aviatrix Controller High Availability (HA)

- Very important: Controller is not in the data path
- If Controller is down → Data Plane still functions
- Your cloud network is still up and running
- Do not compare on-prem to cloud
 - Hardware devices cannot be replaced / software is more flexible
 - Cloud operating models are different
 - Cloud processes are different
 - We need a fresh and different look to solve



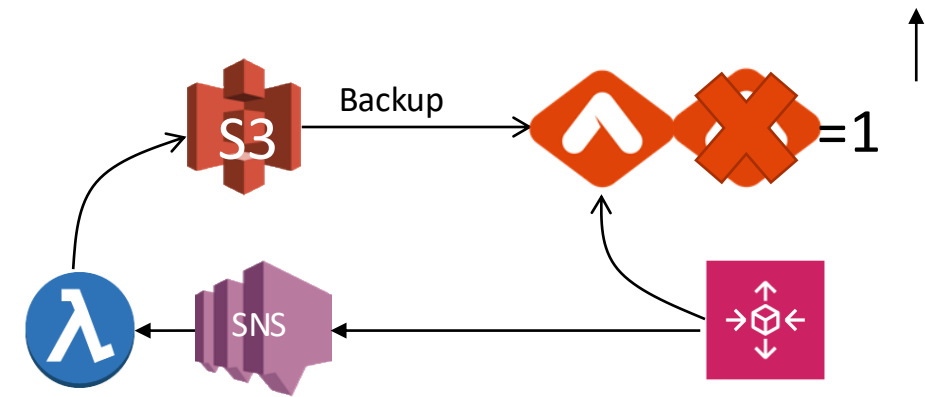
Aviatrix Controller HA Process

- Takes minutes to switch over to new controller
 - Depends on factors such as AWS latency, instance type, size of the DB, etc.
- Previous controller is terminated
- All existing configuration is restored
- New Private IP is assigned (new AZ)
- New controller stays at the same version as previous



Aviatrix Controller HA Process

- Aviatrix Controller HA operates by relying on an AWS Auto Scaling Group
- The Auto Scaling Group has a desired capacity of 1
- If the Controller EC2 instance is stopped or terminated, it will be automatically re-deployed by the Auto Scaling Group
- An AWS Lambda script is notified via SNS when new instances are launched by the Auto Scaling Group
- This script handles configuration restore using the most recent Controller backup file, stored in S3





S2C Best Practices

External Connections – Best Practices

Regional Redundancy:

- Deploy S2C connections across multiple cloud regions to eliminate single points of failure.
- Active-Active HA: Establish multiple tunnels to ensure continuous connectivity and automatic failover if a gateway or tunnel fails.

Intelligent Traffic Engineering:

- BGP AS Path Prepend: Manipulate path selection to precisely influence inbound/outbound traffic flows.
- BGP Route Approval: Enforce security and stability by preventing unauthorized route injections.

Addressing & NAT:

- Resolve IP conflicts between cloud and on-premises environments by utilizing Mapped NAT.

Tunnel Stability:

- Keep-alives: Enable Periodic Ping to prevent security association timeouts during periods of inactivity.
- Fast Convergence: Implement Event-Triggered HA and fine-tune detection timers—especially in overlapping network scenarios—to minimize failover disruption.

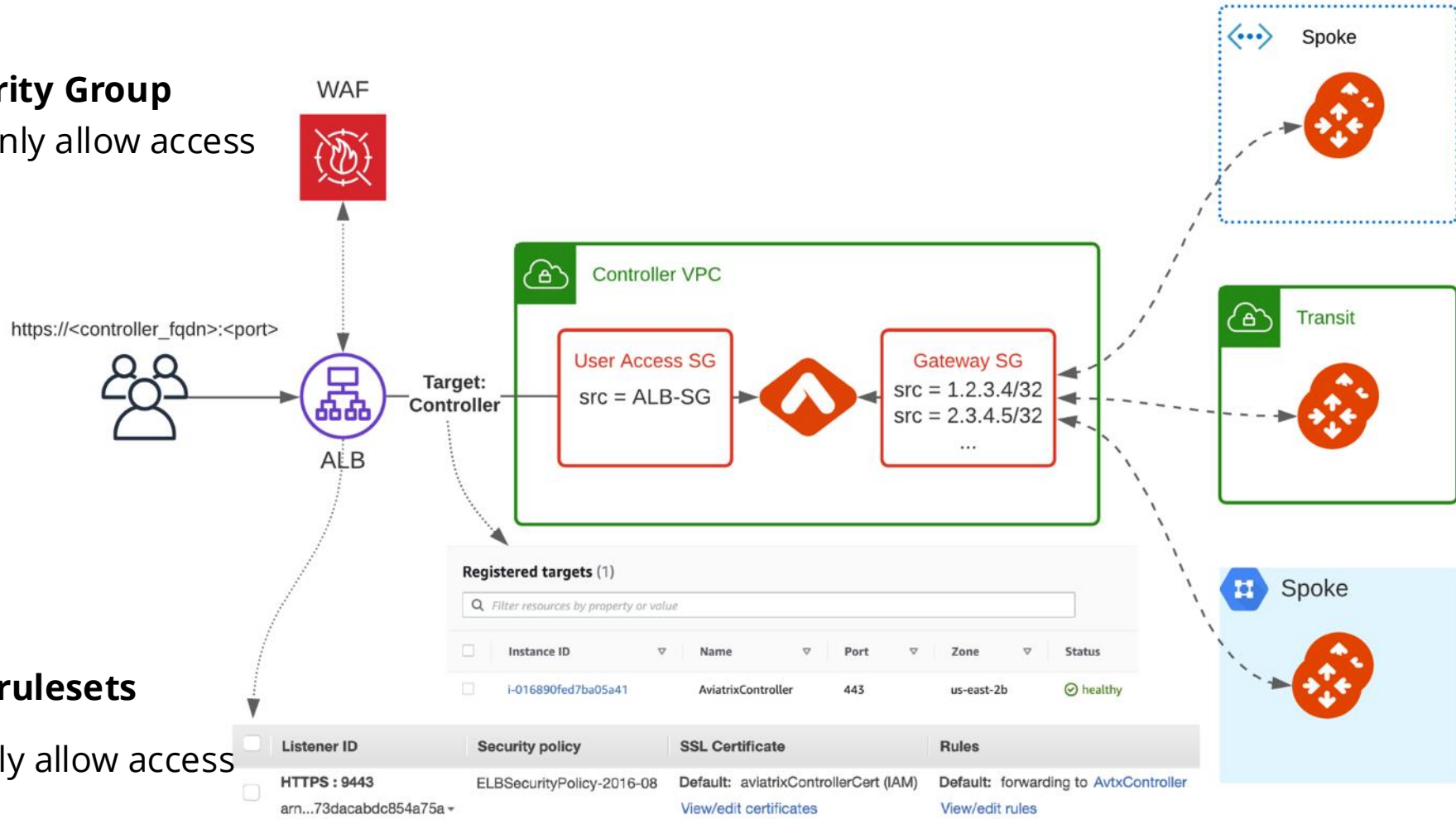


Securing Aviatrix Controller with Application Load Balancer

Applies to any cloud



- Confirm that the **Controller Security Group** Management is NOT disabled to only allow access to the Controller EIP from Aviatrix Gateways
- Create a new internet facing **ALB**
- Modify main **Controller Security Group** to only allow access from the ALB Security Group
- Enable **WAF** on the ALB with AWS Managed Rules
- Adjust ALB **idle timeout**, **modify rulesets**
- Modify **ALB Security Group** to only allow access from the admin user IP





Gateway and Controller Sizing

Controller Sizing

- Controller uses multiple cores to handle the API query load generated by CoPilot → **Minimum 4 core instance**
- Resizing:
 - If you do not use User VPN
 - ❑ Stopping the controller to resize does not impact the data traffic
 - ❑ Always good practice to backup controller before performing upgrade
 - If you use User VPN
 - ❑ No impact to connected users, but new connections could not be established during the stop and resize
- Maintenance Windows for resizing usually do not require more than 15 minutes

Gateway Sizing

- Gateway selection affects expected throughput
- If you decide to enable **High Performance Encryption**
 - Use Jumbo MTU and to verify MTU along the path
 - Go to TROUBLESHOOT > Diagnostics > Network
 - Select a gateway and destination IP address, click **Trace Path**
 - It will display MTU of the devices along the path
- **Secure Egress**
 - T2.micro is not adequate, for instance
 - But test it out and adjust accordingly based on CSP quotas*
 - *CSPs have quotas on PPS, but often do not publish them



Gateway and Controller Upgrading & Updating

Types of Upgrades and Updates

• Software Upgrade

- Replaces relevant Platform (i.e., Controller) and **selected** Gateway packages, configuration files, and binaries to Target version
- Part of regular maintenance operations
- Hitless

• Image Upgrade

- Replaces **selected** Gateway cloud image (AMI, VHD, etc.) to the newer version
- Doesn't change Aviatrix software version
- Less frequent
- Incurs traffic disruption

• Security Patches

- Released when security updates to underlying software components become available.
- Most security patches are hitless (review the release notes)

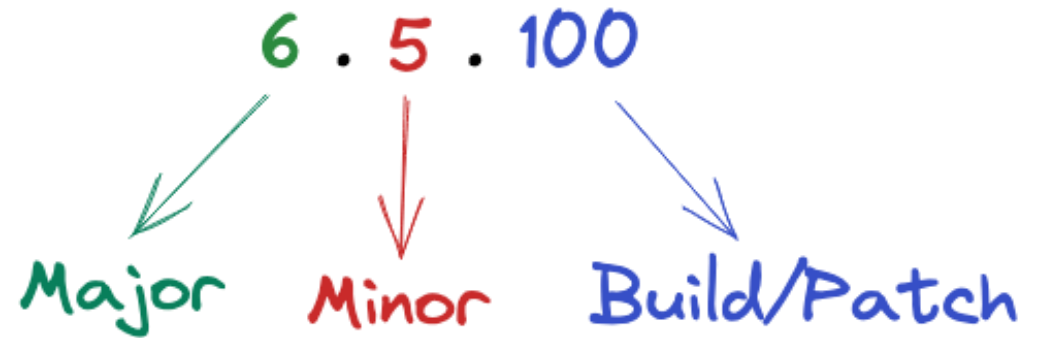
• Software Patches

- Released to address compatibility issues when they arise (if you are using any applications or configurations affected by the patch.
- Most software patches are hitless (review the release notes)

Terminology



- Software Major, Minor, Build Release
 - Numbering convention
 - Example: Aviatrix Release 6.5.100



Supported Upgrade Paths



- Upgrading Builds (within same minor release)

- You automatically get the latest build and cannot select the build number.
- Process might skip over previously released build numbers.



- Upgrading Minor Release Version (within same major release)

- You must upgrade each minor release sequentially.



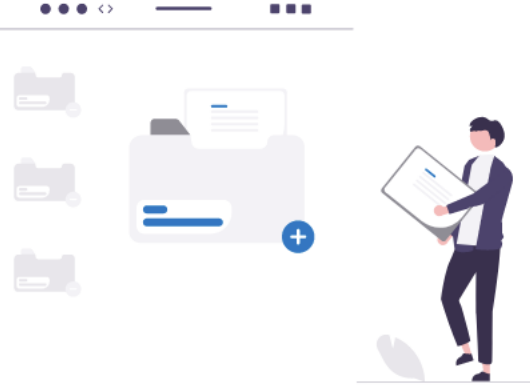
- Upgrading Major Release Version

- You must upgrade each major release sequentially.



Upgrade Groups (1)

- ❑ Each Upgrade Group that gets created is split up into two groups:
 - **Primary Node Group**
 - **Secondary Nodes Group**
- ❑ If you do want to get around that you will have to, after you have created the group, to manually update it to include additional ones. Likewise, If you want to do both primary and Secondary nodes at the same time.



Upgrade Plan and Upgrade Groups

You can now create Upgrade Groups of Gateways based on Cloud, Account, Region, or other criteria.

Each Upgrade Group will be automatically divided into two subgroups: one comprising High Availability (HA) instances and another comprising Primary instances.

Reorder the groups based on your preference to create your Upgrade Plan. Applicable for both Software and Image Upgrades.

Upgrade Groups (2)



☰ CoPilot

Dashboard

Cloud Fabric

Networking

Security

SmartGroups

Cloud Resources

Monitor

Diagnostics

Billing & Cost

Administration

User Access

Reports

Audit

Upgrade

Upgrade

Upgrade Plan

⚠ Upgrade Plan is in Preview. Preview features are not safe for deployment in production environments.

Gateways Requiring Upgrade

Software Upgrade
0

Image Upgrade
0

Prepare For Upgrade

+ Upgrade Group

Prepare For Upgrade

🔍 📄 ⬇

☐	Name	Gateway Type	Gateway Instance	Software Version
☐	Unassigned HA	Transit, Spoke	3	7.0.2239
☐	Unassigned Primary	Transit, Spoke	6	7.0.2239

Controller Version

7.0.2239 Upgrade Controller

Latest Software Version

7.1.3006

Upgrade Gateways in All Upgrade Groups

Target Software Version ⓘ
7.0.2239

Upgrade Group
Unassigned (HA), + 1 More

Gateway Instances
9

☒ Software Upgrade

Gateway software upgrades replace the relevant gateway packages, configuration files, and binaries without disrupting network traffic or replacing the gateways. All software upgrades are hitless.

☐ Image Upgrade

Aviatrix periodically releases new gateway images that include updates, enhancements, and security improvements. A best practice is to plan to upgrade your gateways at least once a quarter.

Software Upgrade Options

Dry Run ⓘ

☒ On

Recommended: On

Software Upgrade

☒ On

Upgrade Groups (3)



Create Upgrade Group

Name

Gateway Selection

Matches all conditions (AND) ▾

Gateway Type ▾ Equal ▾

Spoke x ▾
☐ Transit
☒ Spoke
☐ Edge
☐ VPN
☐ Specialty

Upgrade Group Order
Place Upgrade Group

Bottom x ▾

- ❑ Assign a name to the Upgrade Group
- ❑ Define the **Matching Condition**
- ❑ **Preview Gateways** toggle allows to see the nodes that exactly match the defined condition

Gateway Selection

🔍 Search

Gateway	Gateway Type	Cloud	Region	Account	Tags
AVX-AWS-SPOKE-GW-PROD1-1	spoke	AWS	eu-central-1	aws-account	
AVX-AWS-SPOKE-GW-PROD1	spoke	AWS	eu-central-1	aws-account	
AVX-AZURE-SPOKE-GW-PROD3	spoke	Azure ARM	West Europe	Azure-ACCOUNT	
AVX-AWS-SPOKE-GW-TEST	spoke	AWS	eu-central-1	aws-account	
AVX-AWS-SPOKE-GW-PROD2	spoke	AWS	eu-central-1	aws-account	

Upgrade Groups (4)



+ Upgrade Group										
Prepare For Upgrade										
2 New 2 Modified Discard Commit Search By Upgr ...										
☐	Name	Gateway Type	Gateway Instance	Software Version	Image Version	Kernel Version	Account	Cloud	Region	Upgrade Group Status
☐	Upgrade-Spoke-GW	Spoke	1	7.0.2239	hvm-cloudx-a...	5.4.0-1096-aw...	aws-account	AWS	eu-central-1	Unknown
☐	Upgrade-Spoke-GW	Spoke	4	7.0.2239	[Multiple]	[Multiple]	[Multiple]	[Multiple]	[Multiple]	Unknown
☐	Unassigned HA	Transit	2	7.0.2239	[Multiple]	[Multiple]	[Multiple]	[Multiple]	[Multiple]	
☐	Unassigned Primary	Transit	2	7.0.2239	[Multiple]	[Multiple]	[Multiple]	[Multiple]	[Multiple]	

- ❑ After the Upgrade Group has been created, it will need to be committed before the upgrade gets started
- ❑ The **Edit** icon allows to modify the groups



Support Resources

Support Portal



- Aviatrix customers may visit Support portal – <https://support.aviatrix.com> to access:
 - Knowledge Base with videos
 - Documentation
 - Community
 - History of tickets
 - CSP outage tracker
- Sign up for Email Notifications from Controller

Email Notifications

Manage the status of your Aviatrix system and ensure your teams receive important notification emails sent by Aviatrix.

Enter email aliases for teams that can respond to each type of alert. If you enter the same email for all four fields, that email account could be overwhelmed. [Read more](#)

The email aliases collected will solely be used for the purpose described here. For more information, please refer to our [Privacy Policy](#)

ACCOUNT AND CERTIFICATE ALERTS

Receive important account and certification information.

Administrator Email Alias

ace.lab@aviatrix.com

VERIFY

SECURITY EVENTS

Receive security and CVE (Common Vulnerabilities and Exposures) notification emails.

Security Admin Email Alias

security-admin-group@yourcompany.com

VERIFY

CRITICAL ALERTS

Receive field notices and **critical** notices. These alerts ensure that you can respond to urgent events.

IT Admin Email Alias

it-support@yourcompany.com

VERIFY

STATUS CHANGE NOTIFICATIONS

Receive system/tunnel status notification emails.

IT Admin Email Alias

it-admin-group@yourcompany.com

VERIFY

Status Change Notification Interval (seconds)

60



Next:

Let's troubleshoot together session